
Accessing Seismological Data

1. Data Types

The goal of this lecture is to give you practice downloading seismic and related data. Data collected by projects funded by the various (such as the National Science Foundation) are cleaned up and mostly sent to the IRIS Data Management Center. This isn't necessarily the case for data collected by non-U.S. researchers, and certainly not the case for industry data. Fortunately, there is a huge amount of publically available data out there, and even for data processed and examined, there is almost always something new and different that can be learned from existing data.

But before we try and get data, it's worth considering what exactly we may want to get. Some examples are:

- Data from one earthquake collected at multiple sites near the epicenter, to learn about the near-field effects of the earthquake such as ground acceleration. For this you want high sample rates and seismometers that will not clip at high accelerations.
- Data from one earthquake for a multitude of stations at a range of directions from an earthquake. These can be used to learn about the total moment released by the earthquake, the source-time function of the earthquake, or the focal mechanism.
- A particular phase arrival (like PKP, or P, etc.) from multiple earthquakes within a region that you are exploring. This is useful for methods like shear-wave splitting, tomography, or receiver functions.
- Continuous records to locate earthquakes or perform ambient noise analysis.

Note that non-strong motion seismograms are recorded as **velocity** versus time (to get to ground displacement, you must integrate these numerically; to get to acceleration, you have to take the derivative numerically). Strong motion sensors record acceleration versus time.

Also, there are a lot of metadata available, like:

- Latitude, longitude, or other information about a seismometer station.
- The type of seismometer installed at a given site.
- The length of time data has been recorded
- What sample rate or channels are recorded
- The instrument response to ground motion

And there are time series data collected that are not seismograms:
<http://www.iris.edu/dms/nodes/dmc/data/types/>.

1.1. Data Format

1.1.1. Raw or Archival Formats

Time series data at IRIS are typically stored in a format called Standard for the Exchange of Earthquake Data (SEED). This format is useful for recording as much as is necessary about the seismogram or other sort of time series but is not too useful for actual data analysis. One way

to convert SEED data to a useful format is rdseed (<http://www.iris.edu/dms/nodes/dmc/software/downloads/rdseed/>), although it is more common these days to request the data in a more workable format. miniseed, or mSEED is one more useable subset of the SEED family.

Another format that one may encounter is SEG-Y, which is the Society of Exploration Geophysicists Y Format. This is often how active source data are recorded.

There are also many instrument or digitizer native formats, and legacy formats that you may encounter, but these days I find that not so common. We won't explore it in this lab, but ObsPy, a Python toolbox, is quite good at reading different formats.

1.1.2. Processing-friendly formats

A common format for processing data is .sac, short for Seismic Analysis Code. The format is simple and easy to read into something like Matlab. Furthermore, the Seismic Analysis Code (SAC) itself is an easy-to-use way to process data. I find it useful to do initial processing with SAC, but some people do most of their analysis with SAC, with perhaps a little Fortran code thrown in.

1.2 Channel Formats

Three important parameters to consider in a seismogram are the frequency range the instrument is sensitive to, the sample rate, and the channel being recorded. Let's examine these in a little more detail:

- **Band code** or Frequency band: this is the range of frequencies that can be reasonably found in the data. The sample rate of the data (samples per second) is appropriate to the frequency band. For instance, if the highest frequency signal one is considering is 5 Hz, then a 10Hz sample rate is sufficient (see: http://en.wikipedia.org/wiki/Nyquist_frequency). Typically a structural seismologist would use B (broad band), and an earthquake seismologist may use E, S, or H formats (higher frequencies).
- **Instrument code**: Basic categories of instrument and gain are recorded by these. For teleseismic data one would want to use code H, for a high gain seismometer. Near source seismology may be done with L or N data.
- **Orientation code**: the component of the data being recorded, such as N (north-south), E (east-west), or Z (vertical).

Typically, someone working with teleseismic data would request (BHN, BHE, and BHZ) data, which indicate broadband high gain seismograms in the N, E, and Z orientations. For more information, see Appendix A in the SEED Manual here:

http://www.fdsn.org/pdf/SEEDManual_V2.4.pdf

1.3 Stations and Networks

A station is a location where one or more seismometers and data logging equipment are located. Stations are typically named with three or four character designations, such as ANMO, or Y45.

Networks are sets of stations and are indicated by two-character network codes. There are permanent networks, with unique network codes: <http://www.iris.edu/mda>, and also temporary networks created when a temporary field deployment occurs. These have network codes that are unique to a given time, but not unique over all time. These network codes often start with “X”; for instance, XR indicates a series of temporary networks--only one network may have designation XR at a given time--but more than one network has used the XR code for the thirty or so years we’ve been collecting data. See <http://www.iris.edu/mda#tnetlist> for a list of temporary networks.

2.0 Data Request Tools

2.1 Looking at stations and channels

There are a plethora of data request tools, too many to go into here. To help motivate our use of data, let’s consider an example seismology project that looks at data around the Yellowstone Hotspot, say in a latitude longitude box bounded by -115 and -109 in longitude and 42 and 46 in latitude.

First, let’s see what stations may be useful. Let’s assume we want BHZ data. Plug this into your favorite web browser.

<http://www.iris.edu/gmap/?minlat=42&maxlat=46&minlon=-115&maxlon=-109&chan=BHZ>

Looks like there are a fair amount of stations. Another way to do this is through the **find_channels** command that comes with Standing Order for Data (SOD). You can get this, and other great software, at: <http://www.seis.sc.edu/downloads/>

Since we only want stations with the BHZ channel, we will use the **find_channels** command (this will take a minute or two to complete--hit control-C when you have seen enough of the output)

```
>> find_channels -R -115/-109/42/46 -c BHZ
```

You’ll note the output is a bit complex. Any time a seismometer is changed out at a station, a new record is listed. If we want a list of just the station names, we can do this:

```
>> find_channels -R -115/-109/42/46 -c BHZ | awk '$4 ~ /BHZ/{print $4}' | uniq | cut -d. -f2
```

(Note that you may have to type this in by hand; sometimes copying and pasting doesn’t work with the single quotes.)

Here I tell **awk** to look in field 4 for some text containing “BHZ”. If a line meeting this criteria is found, print that text, pipe it through the **uniq** command which outputs everything but duplicates, and then pipe this to the **cut** command that extracts the actual file name. If you want to get fancy you could next pipe it to the **sort** command to order these alphabetically.

2.2 SOD--the command line method

SOD is a program that takes a different approach to getting seismograms. It considers stations, channels, networks, and events as somewhat separate entities, and gets seismograms that meet all the restrictions you put on the data. One way to approach getting data is through a bunch of command line programs, such as [find_events](#), piped together. So to get the data we have asked for for Yellowstone, we could use these commands:

- **find_events -b 2000 -m 8.0.** This will find all events with magnitudes > 8.0 after Jan. 1, 2000.
- **find_channels -R -115/-109/42/46 -c BHZ.** This will find all channels operating within the Yellowstone latitude longitude box that have the channel BHZ.
- **find_seismograms -B -1ttp -E +2ttp -c BHZ .** This retrieves seismograms for the channel BHZ, and cuts the data at 1 minute before ("-1") the first Pphase arrival ("ttp") and 2 minutes after this arrival ("+2ttp"). This picks any P-phase: P, PP, PKiKP, etc., whichever one arrives first. To change it to just P use ("P"), to change it to the first S arrival use ("tts"), to get seismograms beginning at one minute before the first P arrival and ending one minute after the first S arrival one would use -B -1ttp +E +1tts.

To get seismic data from SOD, one must send SOD a "recipe", which is an XML (a type of markup language) file that tells SOD what sort of data to look for. When you run, say, [find_events](#), you are actually creating a recipe and sending it to sod. To see the recipe that [find_events](#) uses, add the -r flag to the [find_events](#) command.

```
>> find_events -b 2000 -m 8.0 -r > test.xml
```

Now cat test.xml to see the file

```
>> cat test.xml
```

You can see there are a number of "tags" that are giving orders to SOD on what data it should get. One can get the same results by telling SOD itself to use the recipe

```
>> sod -f test.xml
```

We can make a larger recipe that gets seismograms by piping the recipe from [find_events](#) into [find_channels](#), and then piping that into [find_seismograms](#) using the commands we talked about in the bullet points.

```
>> find_events -b 2000 -m 8.0 -r | find_channels -R -115/-109/42/46 -c BHZ -r | find_seismograms -B -1ttp -E +2ttp -c BHZ
```

2.4 SOD--using recipes

Run the command above, but rather than have it get seismograms, get the recipe by adding a -r to the [find_seismograms](#) figure.

```
>> find_events -b 2000 -m 8.0 -r | find_channels -R -115/-  
109/42/46 -c BHZ -r | find_seismograms -B -1ttp -E +2ttp -c BHZ -  
r > test.xml
```

If you look at text.xml you can see some values that you could change to modify what sod returns. As you can see the structure of a SOD recipe is kind of complicated, but there are a number of other tags that one can add to a recipe to further customize the data that is returned beyond what the command line pipes above can do. For instance, sometimes we use a recipe to return the SKS phase, and we add these features;

- make sure there are no nearby phases that aren't SKS
- make sure the SKS has a signal-to-noise level > 2
- rotate the seismograms into the radial/transverse coordinate system before downloading
- add the nearby phase arrivals to the SAC headers (you'll see why this is useful in the SAC lab).

Making one of these recipes is kind of tricky, but the SOD program is really helpful and once you have a recipe that is useful it is typically easy to adapt to get the same sort of data but for a different set of seismometers.

One final word on Sod, `rm -rf ./SodDb` if your recipe didn't run the first time.

2.5 Web Services

By entering in the correctly formatted set of commands to a web address, you can get lists of stations, data, or even synthetic seismograms. Say, for instance, you want to see the travel times of different seismic phases for an event that occurred at (-36.122,-72.898), with a depth of 22.9 km, to a station located at (34.946,-106.457). Just use the following URL.

[https://service.iris.edu/irisws/traveltime/1/query?staloc=\[34.946,-106.457\]&evloc=\[-36.122,-72.898\]&evdepth=22.9](https://service.iris.edu/irisws/traveltime/1/query?staloc=[34.946,-106.457]&evloc=[-36.122,-72.898]&evdepth=22.9)

There are all sorts of amazing web services options, with URL builders to help you get started. Check out: <http://service.iris.edu/>.

If you want to save the results to a web file, just use wget

```
wget -O -  
"http://service.iris.edu/irisws/traveltime/1/query?staloc=[34.946  
, -106.457] &evloc=[-36.122, -72.898] &evdepth=22.9"
```

You could generate these commands with a shell script, perhaps one that reads the station locations from a separate file that you produced from a previous query.

For power users, there are also python, perl, and matlab commands to get data.
<http://service.iris.edu/clients/>

3.0 Homework

1. Use **find_events** and pipes to get SKS-waves that meet these criteria
 - magnitude 7.0 and above
 - BHZ,BHE,BHN channels (which means anything that starts with “BH”) do this with -c BHZ,BHE,BHN.
 - seismogram begins at 1 minute before the first SKS arrival.
 - seismogram ends 2 minutes after the first SKS arrival
 - Use only stations within a box that circumscribes Greenland. You don’t have to be exact with latitudes and longitudes, and your box can contain some land outside of Greenland.
 - Use a start date of June of this year. If that doesn’t retrieve any seismograms, use January of this year.

Note that this may take ten minutes or so. Move this data to a directory called “SKSdata.”